

# The First SUPER-IRNET Workshop

## ~ Rebooting Our In-Person Collaboration ~

**Date:** March 23rd and 24th, 2023 (in Japan time)

**Venue:** National Astronomical Observatory of Japan, Mitaka, Tokyo, Japan

**Room:** Large Seminar Room (1F) in "W1" Building (Subaru Building)

### Workshop Program (ver. 3/17/2023)

(I) : invited talk

(\*) : remote talk

#### Day1 (March 23 in Japan time):

##### [Opening]

9:00-9:05 Opening Remark

Michitoshi Yoshida (NAOJ, Vice Director of General)

9:05-9:25 "Subaru Telescope"

Michitoshi Yoshida (NAOJ) / Satoshi Miyazaki (NAOJ)

9:25-9:35 "Introduction to SUPER-IRNET"

Yusei Koyama (Subaru/NAOJ)

9:35-9:50 "SUPER-IRNET FY2021-FY2022 activity report"

Yusei Koyama (Subaru/NAOJ)

##### [SWIMS session]

9:50-10:05 "SWIMS: Overview and current status"

Masahiro Konishi (Univ. of Tokyo)

10:05-10:15 "Development of SWIMS-IFU"

Kentaro Motohara (NAOJ)

(\*)10:15-10:25 "SWIMS Ruby-Rush"

Ken-ichi Tadaki (NAOJ)

10:25-10:35 "From MAHALO to SWIMS18 surveys"

Kazuki Daikuhara (Tohoku Univ.)

10:35-11:00 (Coffee Break)

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**[ULTIMATE session]**

- 11:00-11:15 “ULTIMATE-Subaru project overview”  
Yosuke Minowa (Subaru/NAOJ) & Celine D’Orgeville/Noelia Martinez Rey (ANU)
- 11:15-11:25 “Subaru collaboration with ASIAA”  
Shiang-Yu Wang (ASIAA)
- 11:25-11:30 “ULTIMATE-Subaru science overview”  
Yusei Koyama (Subaru/NAOJ)
- 11:30-11:40 “Galactic center and bulge science with ULTIMATE”  
Shogo Nishiyama (Miyagi Univ. of Education)
- 11:40-11:55 poster flash talk
- 11:55-13:30 (Lunch + poster)

- (\*)13:30-13:45 (I) “Massive quiescent galaxies at  $z>3$  and the implications for ULTIMATE surveys”  
Karl Glazebrook (Swinburne Univ.)

**[PRIME session]**

- 13:45-14:00 “Status of PRIME NIR telescope”  
Takahiro Sumi (Osaka Univ.)
- 14:00-14:10 “PRIME Microlensing Survey, and Other Sciences in Non-Bulge Seasons”  
Shota Miyazaki (ISAS/JAXA)
- (\*)14:10-14:20 “Mergers, disruptions and disturbances: The explosive facets of binary stellar evolution revealed with infrared time domain surveys”  
Kishalay De (MIT)

**[Euclid session]**

- 14:25-14:45 (I) “Euclid mission pre-launch status”  
Jean-Charles Cuillandre (CEA Paris-Saclay)
- 14:45-15:00 “H20 : Euclid-HSC Deep Field Survey (Potential Roman Deep Fields)”  
Nao Suzuki (Lawrence Berkeley National Laboratory)
- 15:00-15:15 WISHES data production  
Hisanori Furusawa (NAOJ)
- 15:15-15:20 Workshop Photo
- 15:20-15:40 (Coffee Break)

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**[Roman session]**

15:40-16:00 (I) “Status of the Nancy Grace Roman Space Telescope”

Julie McEnery (NASA/GSFC)

16:00-16:10 “Photo-z calibration for Weak Lensing Cosmology with Nancy Grace Roman Space Telescope”

Hironao Miyatake (Nagoya Univ.)

16:10-16:20 “Time-domain Science with Nancy Grace Roman Space Telescope”

Takashi Moriya (NAOJ)

16:20-16:30 “Input into the Roman's Galactic Bulge Time Domain Survey”

Daisuke Suzuki (Osaka Univ.)

(\*)16:30-16:40 “Observing High Redshift Universe with Roman and Its Synergy with Subaru”

Yuichi Harikane (Univ. of Tokyo)

16:40-17:00 Discussion on the future of SUPER-IRNET

**Day2 (March 24 in Japan time):****[Science Talk Session (1)]**

09:00-09:15 “The host galaxies of Fast Radio Bursts”

Stuart Ryder (Macquarie Univ.)

09:15-09:30 “Understanding galaxy quenching through statistical study of RQGs”

Zhiying Mao (Tohoku Univ.)

09:30-09:45 “Evolution of protoclusters across cosmic time studied with infrared all-sky surveys”

Mariko Kubo (Tohoku Univ.)

(\*)09:45-10:00 “The mid- + far-IR probe project PRIMA for the early 2030”

Denis Burgarella (Laboratoire d'Astrophysique de Marseille)

10:00-10:15 “GREX-PLUS”

Akio Inoue (Waseda Univ.)

10:15-10:45 (Coffee Break)

**[Special Session on Diversity ]**

10:45-11:45 (I) Special Talk 1 “What is holding female scientists back?”

Hiromi Yokoyama (Univ. of Tokyo)

11:45-13:00 (Lunch + poster)

**[Science Talk Session (2)]**

13:00-13:15 “Perturbation theory challenge in redshift space accelerated with response function approach”

Ken Osato (Chiba Univ.)

13:15-13:30 “BAO mock measurement of the 3D correlation function for photometric surveys”

Keitaro Ishikawa (Nagoya Univ.)

13:30-13:45 “A study on the relation between formation history and observables of galaxy clusters”

Seongwhan Yoon (Nagoya Univ.)

13:45-14:00 “Testing general relativity with the joint analysis of weak lensing and galaxy clustering from HSC-Y3 and BOSS”

Koki Tanida (Nagoya Univ.)

14:00-14:30 (Coffee Break)

**[Special Session on Future Japanese OIR projects (in Japanese)]**

14:30-15:30 (I) Special Talk 2 “Building OIR Roadmap: History since 2000 and Prospects for the Future”

Toru Yamada (ISAS/JAXA)

15:30-16:10 (I) Special Talk 3 “Japanese Large Projects in Optical and Near-Infrared Astronomy: Current Status and Challenges”

Masami Ouchi (NAOJ/Univ. of Tokyo)

16:10-16:55 Discussion

16:55-17:00 Closing Remark



## Posters:

- Abdurrahman Naufal (SOKENDAI) "*Rest-frame UV morphology of star-forming galaxies at  $z \sim 2$ : Environmental dependence?*"
- Kosuke Takahashi (Tohoku Univ.) "*Ruby-Rush: The quest of red monsters lurking in the protoclusters at  $z \sim 5$* "
- Takahiro Kanai (Saitama Univ.) "*A Deep NIR Survey of Very-low mass Objects in the R CrA Region*"
- Masayuki Akiyama (Tohoku Univ.) "*ULTIMATE-START : Subaru Tomographic Adaptive optics Research experiment*"
- Ronaldo Laishram (Tohoku Univ.) "*Environmental dependence of star-forming galaxies at  $z \sim 0.4$  in the DEEP2-3 Field*"
- Naoki Matsumoto (Tohoku Univ.) "*MIR Search of Heavily Obscured  $z > 3$  AGN Missed in X-ray Surveys*"
- Jose Manuel Perez-Martinez (Tohoku Univ.) "*Coevolution of protoclusters and their galaxy populations at the cosmic noon: The case of PKS1138 and USS1558*"
- Kohei Shibata (Ehime Univ.) "*Environment of SDSS quasars at  $z = 0.3 - 1.0$  traced by Subaru HSC*"
- Bovornpratch Vijarnwannaluk (Tohoku Univ.) "*The Population of Luminous Obscured AGN at Cosmic Noon Unveiled with Deep and Wide Multiwavelength Surveys*"
- Yoshito Ono (Subaru Telescope) "*ULTIMATE-Subaru : GLAO preliminary design overview*"
- Chihiro Tokoku (NAOJ) "*NINJA : Subaru Optical-to-NIR spectrograph optimized for LTAO*"